

MACHINE LEARNING–BASED PREDICTION OF DIGITAL PAYMENT ADOPTION AMONG RETAIL VENDORS

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ABSTRACT

The adoption of digital payment platforms by retail vendors has become a pivotal transformation in India's payment landscape. This project utilizes machine learning techniques to predict whether a retail vendor is likely to adopt digital payment systems such as Paytm, Google Pay, PhonePe, and Amazon Pay. Data was collected from 270 merchants across 10 major retail markets in Bangalore, targeting both organized and unorganized sectors. Through a comprehensive analytical approach involving descriptive statistics, exploratory factor analysis (EFA), and Support Vector Machine (SVM) modeling, the system identifies core factors influencing adoption: perceived usefulness, social influence, and compatibility. Additionally, the system evaluates customer satisfaction across platforms and forecasts future digital payment adoption trends using the ARIMA time series model. This integrated platform consists of three major modules: user (vendor interaction), admin (data management and analysis), and machine learning (prediction and forecasting). The system not only aids in understanding the behavior of retail vendors but also empowers policymakers, fintech providers, and entrepreneurs with actionable insights to improve the digital payments ecosystem.

Keywords: Support Vector Machine (SVM), ARIMA, Exploratory Factor Analysis (EFA)